

Your grade: 100%

Next item →

Learning Check - Recursive Splitting

1. For Questions 1-7, we apply **Learning Check** to fit a quadratic polynomial to a given dataset. Recall that

COACH

Ready to review that you've learned before starting the assignment? I'm here to help.

$$\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y$$
, where $\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix}$

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In what follows, given a dataset $\{(x_i, y_i)\}_{i=1}^n = \{(1, 2), (2, 5), (3, 13), (4, 20)\}$, we will take steps to find the coefficients of a quadratic model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$.

Submitted Sep 30, 11:59 PM IST 4 left (5 attempts every 24 hours)

Try again

Step 1: What is M_X^T ? Choose the correct options below.

- ☐ $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 4 & 9 & 16 \\ 1 & 2 & 3 & 4 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 2 \\ 1 & 9 & 3 \\ 1 & 16 & 4 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}$
- ☒ $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \end{bmatrix}$

Correct

2. What is $M_X^T M_X$?

1 / 1 point

- ☐ $\begin{bmatrix} 30 & 15 & 16 \\ 10 & 30 & 10 \\ 1 & 1 & 30 \end{bmatrix}$
- ☐ $\begin{bmatrix} 4 & 30 & 10 \\ 30 & 354 & 100 \\ 10 & 100 & 30 \end{bmatrix}$
- ☒ $\begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Correct

3. What is $M_X^T y$?

1 / 1 point

- ☐ $\begin{bmatrix} 40 \\ 459 \\ 131 \end{bmatrix}$
- ☐ $\begin{bmatrix} 131 \\ 40 \\ 459 \end{bmatrix}$
- ☐ $\begin{bmatrix} 459 \\ 40 \\ 131 \end{bmatrix}$
- ☒ $\begin{bmatrix} 40 \\ 131 \\ 459 \end{bmatrix}$

Correct

4. What is $\hat{\beta}_0$? (Answer as a real number with one decimal point.)

1 / 1 point

-0.5

Correct

$$\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y = [-0.5, 1.2, 1]^T.$$

5. What is $\hat{\beta}_1$? (Answer as a real number with one decimal point.)

1 / 1 point

1.2

Correct

$$\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y = [-0.5, 1.2, 1]^T.$$
 The second element in this vector is $\hat{\beta}_1$.

6. What is $\hat{\beta}_2$? (Answer as a real number with one decimal point.)

1 / 1 point

1

Correct

$$\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y = [-0.5, 1.2, 1]^T.$$
 The third element in this vector is $\hat{\beta}_2$.

7. What would be your prediction $\hat{y}$ for $X = 2$?

1 / 1 point

☐ 25

☒ 5.9

☐ 5

☐ 2

Correct

The right answer is $-0.5 + 1.2 \times 2 + 1 \times 2^2 = 5.9$.